31/50

**Exam 1 - CY1010 course (50 marks)**

Date: 04.09.2024 Time: 6:30 pm – 8:00 pm

Student name:

Roll No:

Department:

1. The source of second-generation biofuel is
   a) Starch b) Lipids c) Algae d) Rice husk

Ans: ~~(b)~~ (d)

2. The order of $CO_2$, $N_2O$, $CH_4$, $SF_6$ in terms of GWP is
   a) $CO_2 < N_2O < CH_4 < SF_6$ b) $CO_2 < CH_4 < N_2O < SF_6$
   c) $N_2O < CO_2 < CH_4 < SF_6$ d) $CH_4 < N_2O < CO_2 < SF_6$

Ans: (b)

3. What odorizer detects gas leakage in the CNG/LPG cylinders?
   a) $H_2S$ b) Mercaptan c) Mercaptene d) $SF_6$

Ans: (b)

4. Carbon particulate matter absorbs IR radiation because it contains
   a) Adsorbed sulphate b) Adsorbed water c) Adsorbed hydrocarbons d) All of these

Ans: (b)

5. Earth core contains major amounts of
   a) Mo and Fe b) Fe and Ni c) Ni and Mo d) Ni and Cu

Ans: (b)

6. Which chemical production process consumes most of the $CO_2$ industrially?
   a) Salicylic acid b) Urea c) Polycarbonates d) Methanol

Ans: (b)

7. Green hydrogen production
   a) Water splitting b) Electricity is needed c) Catalyst is needed d) All of these

Ans: (d)

8. Which statement is correct about methane emissions during rice cultivation?
   a) Bacterial decomposition of organic matter under aerobic conditions
   b) Bacterial decomposition of organic matter under anaerobic conditions
   c) Methane emissions are not possible during rice cultivation
   d) Both a and b

Ans: (b)

9. What is the intermediate nucleus in nuclear fission of $^{235}U$?
   a) $^{236}U$ b) $^{238}U$ c) $^{144}Ba$ d) $^{89}Kr$

Ans: (a)

10. The main components of LPG are
    a) Propene and butane b) Propane and butane c) Propane and butene d) Propene and butene

Ans: (b)

11. Biodiesel can be produced from
    a) Algae b) Waste cooking oil c) Fat d) All of these

Ans: (d)

12. Which is the stable nucleus?
    a) $^{56}Fe$ b) $^{235}U$ c) $^{238}U$ d) $^{2}H$

Ans: (b)

13. Which statement is correct?
   a) Gasoline engines emit more $SO_2$ than diesel engines
   b) Diesel engines emit more $SO_2$ than gasoline engines
   c) Gasoline and diesel engines do not emit $SO_2$
   d) Both gasoline and diesel engines emit similar amount of $SO_2$

X Ans: (a)

14. Catalyst is used in
   a) Fuel cell   b) Lithium batteries   c) both a and b   d) None of these

X (a)

15. Which among the following compounds has the highest lifetime/residence time in the atmosphere?
   a) Nitrous oxide   b) Sulphur hexafluoride   c) HFC   d) Methane

✓ Ans: (b)

16. Which fuel emits higher carbon emissions?
   a) LPG   b) CNG   c) Petrol   d) Biodiesel

X Ans: (c)

17. Which statement is correct in the solvent-based direct air-capturing method
   a) KOH is regenerated by the reaction of $K_2CO_3$ with $Ca(OH)_2$
   b) $Ca(OH)_2$ is regenerated by the reaction of CaO with $H_2O$, which is essential for DAC method
   c) Regeneration of $Ca(OH)_2$ is not required   d) Both a and b

✓ Ans: (d)

18. The best plastic waste for graphene synthesis is
   a) Polycarbonates   b) PET   c) Polyolefins   d) All of these

X Ans: (a)

19. Biodiesel can be produced via transesterification of
   a) Lipids   b) Algae   c) Lignin   d) Both a and b

X Ans: (c)

20. Bioethanol is produced via fermentation of
   a) Lignocellulose   b) Starch   c) Lipids   d) All of these

X Ans: (a)

21. What is the best plastic to produce jet fuel
   a) PET   b) Polycarbonates   c) Polyolefins   d) Polystyrene

✓ Ans: (c)

22. Amine-functionalized MOFs can adsorb $CO_2$ from air through
   a) Physisorption only   b) Chemisorption only   c) Both chemisorption and physisorption   d) None of these

✓ Ans: (c)

23. Which of the following molecules are Infrared active?
   a) CO   b) $CO_2$   c) $O_2$   d) both a and b

✓ Ans: (d)

24. $N_2O$ emission sources are
   a) Urea   b) Vehicles   c) Coal   d) All of these

X Ans: (a)

25. Which type of water exhibits higher electrical conductivity?
   a) Surface water   b) Irrigation water   c) Saline inland lakes water   d) Drinking water

✓ Ans: (c)

26. Half-life of persistent organic pollutants varies with concentration in
   a) Linear manner   b) Independent   c) Inverse manner   d) Exponential manner

X Ans: (a)

27. Lindane is an organic chemical used as

a) Detergent b) Insecticide c) Making plastic material d) Plasticizer

Ans: (d)

28. The rubber used for manufacturing tires contains

a) Latex b) Cellulose c) Styrene-butadiene d) Nylon-66

Ans: (c)

29. Which one of the following is not a synthetic pesticide

a) Endosulfan
b) Perfluorooctane sulfonic acid
c) Nicotine sulfate
d) Benzoyl sulfonate

Ans: (c)

30. DDT is a popular pesticide and fumigating chemical for killing malaria causing mosquitos. How many carbon-chlorine bonds it has

a) 2 b) 3 c) 4 d) 5

Ans: (b)

31. From 1950 to 2015, primary global production of virgin plastic is 8.3 billion metric tons. How many tons (billion metric) were deployed for usage for a few decades

a) 2500 b) 2.5 c) 4.9 d) 4900

Ans: (b)

32. Plastic waste is currently generated at a rate approaching 400 Mt year$^{-1}$. If it continues at the same rate of production and waste generation, how much primary plastic waste would be generated by 2050?

a) 25000 Mt b) 12000 Mt c) 1800 Mt d) 10000 Mt

Ans: (d)

33. Which of the following organic chemical does not contain carbon-chlorine bonds in the molecular structure?

a) Dioxins b) Endrin c) Dieldrin d) Phenanthrene

Ans: (d)

34. What is the use of Furans?

a) Insecticide
b) pest control
c) both a and b
d) No use

Ans: (c)

35. The molecular mass of monomers is in the range of

a) 10000 – 100000 g
b) 10 – 100 kg
c) 10 – 100 Da
d) 10000 – 100000 Da

Ans: (b)

36. Which of the following polymer is quickly discarded as a waste

a) Poly(vinylalcohol)
b) Polyethylene
c) poly(methylmethacrylate)
d) poly(vinylchloride)

Ans: (b)

37. Degradation rate of plastics follow pseudo-zero order kinetics and half-life varies according to

a) Linear variation with the amount of plastic
b) Independent on the amount of plastic
c) Inverse variation with the amount of plastic
d) Ratio of amount of plastic to rate constant

Ans: (a)

38. To be persistent in the environment, half-life of a chemical in the air must be.

a) ~ half day b) ≥ 48 hours c) ≥ 24 hours d) ≥72 hours

Ans: (b)

39. Which one of the following statements is not correct
   a) Type 5 plastic is safer to produce, and use compared to type 4 plastic
   b) Type 5 plastic is not recommended for recycling
   c) Type 5 plastic is easier to recycle compared to type 1 plastic
   d) Type 5 plastic recycling doesnot require any energy

X

Ans: (b)

40. Which of the following is the least preferred way to reduce plastic pollution?
   a) Dilution   b) Reduce   c) Reuse   d) Incineration

Ans: (d)

41. How much percentage of plastic waste remains in the environment at global level as per the global analysis statistics (1950-2015)?
   a) 9%   b) 69%   c) 89%   d) 79%

Ans: (d)

42. What is the function of scrubbers in modern incinerators?
   a) Separates metals
   b) Regulates particulate matter
   c) Supplies energy to the steam boiler
   d) prevents the emission of toxic gases

Ans: (B)

43. Cleansing action of soaps in the hard water is inefficient due to
   a) Weakly acidic   b) Hydrophobic nature   c) Reduction of surface tension   d) Micelle formation

X

Ans: (d)

44. How much percentage of algal blooms formation in the lakes is caused by the usage of detergents
   a) 75%   b) 25%   c) 50%   d) 15%

Ans: (b)

X

45. The structure and function of biopolymers is due to
   a) secondary structure   b) tertiary structure   c) quaternary structure   d) primary structure

Ans: (d)

46. A large value of Henry constant ($K_H$) of a chemical is an indication for tendency to
   a) Dissolution in alcohols   b) Dissolution in lipids   c) Mixing in water   d) Evaporation from water

Ans: (d)

47. Electron activity, pE, of water is around 25 and it favors
   a) Reduction   b) oxidation   c) Both a and b   d) None

Ans: (b)

48. poly(E-caprolactone) is synthesized through
   a) Chain polymerization
   b) Crosslinking polymerization
   c) Condensation polymerization
   d) Co-polymerization

Ans: (a)

49. Which of the following method estimates accurate amount (relatively) of organic pollutants in the water
   a) Biochemical oxygen demand
   b) Chemical oxygen demand
   c) Both a and b
   d) Electrochemical oxygen demand

Ans: (a)

50. Which of the following chemical gets produced by burning chlorine containing medical and municipal wastes
   a) Dichlorodiphenyltrichloroethane   b) Tetrachlorodibenzo-*p*-dioxin   c) Polychlorobiphenyls   d) 1,4-dichlorobenzene

Ans: (b)

1. D
2. B
3. B
4. D
5. B
6. B
7. D
8. B
9. A
10. B
11. D
12. A
13. B
14. C
15. B
16. D
17. D
18. C
19. D
20. B
21. C
22. C
23. D
24. D
25. C
26. C
27. B
28. C
29. C
30. D
31. B
32. A
33. D
34. D
35. C
36. B
37. D
38. B
39. C
40. D
41. D
42. B
43. A
44. B
45. C
46. D
47. B
48. A
49. A
50. B